

Sourasis Karak

Kalyani, West Bengal, India

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Professional Summary

Second-year Computer Science undergraduate focused on Machine Learning and Data Science, with specialized interests in predictive modeling, distributed systems, and computational biology. Experienced in designing leakage-free validation frameworks, engineering robust ML ensembles, and architecting resilient backend data pipelines.

Education

JIS College of Engineering

Kalyani, West Bengal

B.Tech in Computer Science & Engineering, Current: 2nd Year

2025–2029

Coursework: Data Structures & Algorithms, DBMS, Linear Algebra, Probability & Statistics, Object-Oriented Programming.

Technical Projects

github.com/thekarak/Analyticus

Team Leader | PlayHack ML Track, IIT Guwahati

Analyticus — Athlete Injury Risk & Recovery Prediction

2026

- Designed a leakage-free validation pipeline with grouped 5-fold cross-validation and fold-only-fit preprocessing, ensuring zero athlete overlap between train and test splits.
- Engineered a dual-stage ensemble (XGBoost, LightGBM, CatBoost) over 77 biometric and activity features, achieving an out-of-fold F1-score of 0.9995 and 100% recall with zero false negatives.
- Calibrated decision threshold from 0.5 to 0.12 to counter strict competition false-negative penalties while preserving classification precision.

github.com/thekarak/VARUNA

Smart India Hackathon 2026

VARUNA — Maritime Oil-Spill & Vessel Investigation System

2026

- Formulated end-to-end technical system architecture fusing satellite SAR/optical imagery, AIS vessel telemetry, and ocean-drift simulation for maritime forensic investigation.
- Designed core ML pipeline combining ESRGAN super-resolution with U-Net segmentation for precise oil-slick boundary delineation and drift trajectory tracking.
- Engineered scalable backend computing infrastructure utilizing FastAPI, Celery asynchronous workers, Redis task queues, and PostGIS for spatial telemetry queries.

roastmebuddy.vercel.app

Solo Developer | Production Deployment

RoastMeBuddy — AI-Powered Resume Feedback Platform

2026

- Built and shipped a responsive web application delivering structured, context-aware feedback on technical CVs using LLM APIs.
- Implemented prompt engineering with strict JSON output schemas to guarantee deterministic scoring and parsing across heterogeneous resume formats.
- Deployed the complete product live on Vercel with asynchronous API handling and low-latency response streaming.

Technical Skills

Programming Languages: Python, C, SQL, HTML5, CSS3

Machine Learning & Data: XGBoost, LightGBM, CatBoost, Scikit-Learn, Ensemble Modeling, Feature Engineering, K-Fold CV, Pandas, NumPy

Deep Learning & Backend: ESRGAN, U-Net Segmentation, FastAPI, Celery, Redis, PostGIS, RESTful APIs

Developer Tools & DevOps: Git, GitHub, Linux/Bash, Streamlit, Vercel, Microservices Architecture

Certifications & Leadership

Certifications: Solution Architecture Job Simulation — Forage (Virtual Experience, 2026): Distributed cloud systems and architecture design.

Leadership: Active Technical Member, College Technical Club (2025–Present): Lead competitive programming and peer developer workshops.

Community: Extended Member, devdotcom.in (2025–Present): Participate in open-source discussions and collaborative software projects.